

FINAL EXPLANATION: PHYSICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: PHYSICS GRADE 10 – SUB-STRAND 3.3 INTRODUCTION TO ELECTRONICS: FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Driving question: Why does a silicon solar panel on a Kajiado farm produce less power in the hot afternoon sun (about 35 °C) than on a cool morning (about 18 °C), and what is so special about silicon that it can act as both a conductor and an insulator depending on the conditions? Use everything you learned in Lessons 1 to 7: your circuit tests, band diagrams, doping diagrams, the PhET simulation and the case studies. Answer each section in the blank space under its prompt. Write in full sentences. Use the scientific words we learned (valence band, conduction band, band gap, electron-hole pair, doping, n-type, p-type, p-n junction). Where a prompt asks for a diagram, draw it neatly and label it. Where a prompt asks for evidence, refer to specific results from our lessons.

Part 1: What our circuit tests showed

In Lesson 1 you tested materials in a 1.5 V circuit with an LED. Here are the results:

Material	LED brightness
Copper wire	Fully bright
Steel nail	Fully bright
Plastic sufuria handle	Off
Rubber eraser	Off
Graphite pencil lead	Dim
Silicon chip	Dim

(a) Conductors: copper wire and steel nail, because the LED was fully bright. Insulators: plastic sufuria handle and rubber eraser, because the LED stayed off. Partial conductors: graphite pencil lead and silicon chip, because the LED glowed only dimly. (b) Materials that conduct only partially are called semiconductors. Silicon is one, because its conductivity is between that of a conductor such as copper and that of an insulator such as plastic. Its conductivity also changes with conditions such as light and temperature. (c) Ohm's Law and our simple-circuit ideas describe how current flows through a material when a voltage is applied, but they do not explain why a material conducts a little, or why its conductivity changes with light or temperature. They also could not explain why the panel gives no power in darkness, or why heating it makes it work worse. We needed a new model of how electrons are arranged inside solids, the energy band model.

(a) Use the table to sort the materials into conductors, insulators and materials that conduct only partially, and give your reason for each group. (b) Name the category of materials that conduct only partially, and explain why a material in that category has its name. (c) Why could your earlier ideas about simple circuits and Ohm's Law not explain the silicon result or the Kajiado solar panel?

Part 2: The energy band model and silicon's dual identity

(a) Draw and label simple energy band diagrams for a metal such as copper, an insulator such as the plastic handle or rubber, and a semiconductor such as silicon. Show the valence band, the conduction band and the band gap on each. (b) Explain how the size of the band gap decides whether a material is a conductor, an insulator or a semiconductor. (c) Silicon's band gap is about 1.1 eV. Use this to explain why silicon can behave like an insulator in darkness but conduct when sunlight falls on it. Then explain what happens to the number of electron-hole pairs in pure silicon when it is heated, and why this is a different question from how well a solar panel performs.

(a) Copper: the valence band and conduction band overlap, so there is no band gap and electrons move freely. Plastic or rubber: the valence band is full and a very large band gap separates it from an empty conduction band. Silicon: the valence band is full and the conduction band is empty at low energy, with a small band gap of about 1.1 eV between them. Each diagram labels the valence band (where electrons are normally held), the conduction band (where electrons can move freely) and the gap. (b) Electrons only carry current when they are in the conduction band. In a metal there is no gap, so electrons are always free and the material conducts well. In an insulator the gap is so large that electrons cannot cross it at normal conditions, so almost no current flows. In a semiconductor the gap is small, so a few electrons can cross if they receive energy. (c) In darkness, the electrons in silicon have too little energy to cross the 1.1 eV gap, so almost no electrons reach the conduction band and silicon behaves like an insulator. When sunlight supplies enough energy, electrons are pushed across the gap into the conduction band. Each one leaves behind a hole in the valence band, which makes an electron-hole pair. These carriers can flow as current. This is why a silicon panel produces electricity in light but not in darkness. In pure silicon, heating also promotes more electrons across the gap, so more electron-hole pairs form. But a solar panel's output depends on how many light-excited electrons complete the external circuit through the p-n junction, not just on how many carriers exist. Heat also increases lattice vibrations, so the effect of temperature on the panel's output has to be explained separately, as we do in Part 4.

Part 3: Doping and the p-n junction

The silicon in a real solar panel is not pure. (a) What is doping, and why do engineers do it? (b) Explain how phosphorus makes n-type silicon and how boron makes p-type silicon. Say which charge carriers are the majority in each. (c) Describe what a p-n junction does to the flow of charge, using what you saw in the PhET simulation about forward bias and reverse bias. (d) Explain why the p-n junction matters for the Kajiado solar panel.	(a) Doping is the deliberate addition of a small amount of impurity atoms to pure silicon to control its conductivity. Doped silicon has a lower resistivity than pure silicon and the higher the doping concentration the lower the resistivity, so it conducts much better and in a controlled way. (b) Phosphorus (Group V) has one more outer electron than silicon needs for bonding. This extra electron is free to move, so phosphorus is a donor and makes n-type silicon, where electrons are the majority charge carriers. Boron (Group III) has one fewer outer electron, which leaves a hole, a missing electron. This makes p-type silicon, where holes are the majority charge carriers. (c) Where p-type and n-type silicon meet, a depletion region forms with a built-in electric field that acts as a barrier. In forward bias the applied voltage lowers this barrier and a large current flows. In reverse bias the barrier is raised and almost no current flows. So the I-V curve is asymmetric and the junction lets current go in one direction only. (d) The panel's silicon is a carefully made combination of n-type and p-type regions. When light excites electrons across the band gap, the junction drives them in one direction through the external circuit as useful current, for example to the farmer's water pump. Without the junction, the electrons would not be pushed in a single direction.
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Part 4: Solving the Kajiado solar panel mystery	
Mama Akinyi's silicon panel in Kajiado gives less power at about 35 °C on a hot afternoon than at about 18 °C on a cool morning, even though the afternoon sun is strong. Write a Claim–Evidence–Reasoning explanation. (a) Claim: answer the driving question in one or two sentences. (b) Evidence: give at least two pieces of evidence from our lessons (for example the temperature data, the band diagrams, the PhET simulation or the case study graph). Say what each piece shows and be careful about how closely its temperature range matches the Kajiado panel. (c) Reasoning: use the energy band model, the p-n junction and electron-hole pairs to explain how a higher temperature leads to your claim. (d) Explain why the claim 'more heat means more electricity' does not hold for this panel, and why it matters	(a) Claim: The panel gives less power in the hot afternoon not because there is less light, but because the higher temperature (about 35 °C) increases lattice vibrations in the silicon crystal, which raises electron-hole recombination in the p-n junction, so fewer light-excited electrons complete the external circuit than on the cool morning (about 18 °C). (b) Evidence: The temperature-vs-power graph for a silicon panel based on the Kajiado farm shows power falling as the temperature rises from 18 °C to 35 °C. The case study graph for a silicon panel in Lodwar shows the same trend over a wider range, 15 °C to 45 °C, which supports the pattern but is a different place and not the Kajiado data. The band diagrams show that silicon's small gap (about 1.1 eV) lets both light and heat promote electrons. The PhET simulation showed that heating the doped silicon produced more thermally generated electron-hole pairs and weakened the junction's built-in voltage; it covered a much wider range (200 K to 400 K) than the 17 °C difference in Kajiado, so it shows the direction of the effect rather than the exact size. (c) Reasoning: Light gives electrons the energy to cross the band gap, and the p-n junction then pushes them in one direction round the circuit. At a higher temperature the atoms of the silicon crystal vibrate more strongly. These lattice vibrations raise the rate of electron-hole recombination, so more electrons and holes meet and cancel before the electrons can do useful work. The vibrations also scatter the carriers. Fewer electrons therefore complete the external circuit through the pump. Heat also generates extra electron-hole pairs, which weakens the junction's built-in voltage, so the panel's output voltage drops as well. On the cooler morning there is less recombination, so more photon-excited electrons cross the junction and flow through the circuit, and the efficiency is higher. (d) A simple resistor only follows Ohm's Law, but a semiconductor's behaviour depends on band gap, carriers and the junction, so extra heat can make the panel work worse. Silicon works because its intermediate band gap lets sunlight excite electrons, and doping makes a p-n junction that sends them one way. Copper has no gap and no junction effect, and rubber has a gap that is too large, so neither could work as a solar panel material.

that silicon is neither copper nor rubber.	
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Part 5: Semiconductors in Kenyan life	
(a) Name two Kenyan technologies, other than the Kajiado panel, that depend on semiconductors. For each one, say which semiconductor idea from our lessons it uses (band gap, doping, p-n junction or temperature effects). (b) Explain how understanding and engineering semiconductor materials has changed everyday life in Kenya. (c) In which way did your thinking change between Lesson 1 and now?	(a) Examples: (1) The microchips inside M-PESA mobile handsets contain doped p-n junctions that act as transistor switches, allowing one-way electron flow and digital logic at microscopic scale. (2) M-KOPA solar home systems use silicon panels that rely on the band gap and the p-n junction to turn light into current. KPLC smart meters and Safaricom network chips also depend on doped semiconductors. (b) By controlling band gaps and doping, engineers can make tiny switches and light-to-electricity converters. This has brought solar power to rural places such as schools in Turkana and farms in Kajiado, made mobile money possible through M-PESA, and made electricity use easier to measure with smart meters. These devices are small, cheap and reliable because semiconductors can be engineered precisely. (c) In Lesson 1 most of us thought that more heat and more sunlight must always mean more electricity. Now we know that in a semiconductor the temperature affects the carriers and the junction, so a hot panel can give less power even in bright sunshine. We also now understand that silicon is special because of its small band gap and because doping lets us build junctions, not simply because it is a conductor or an insulator.

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of the energy band model	Correctly describes valence band, conduction band and band gap for a metal, an insulator and silicon (about 1.1 eV), and uses them to explain why silicon conducts in light but not in darkness.	Describes the bands and gaps for most materials and explains silicon's conduction, with small gaps in detail or vocabulary.	Mentions bands or gaps but confuses which material has which gap, or does not link the gap to conduction.
Explanation of doping and the p-n junction	Accurately explains n-type (phosphorus, electron donor) and p-type (boron, holes), the majority carriers, and how the junction gives one-way current, linked to the solar panel.	Explains n-type and p-type correctly and mentions one-way flow, but the link to the panel is brief.	Confuses n-type and p-type or the carriers, or does not explain the junction's role.
Claim–Evidence–Reasoning on the Kajiado phenomenon	Gives a clear claim, at least two specific pieces of evidence from the lessons (noting how well each matches the Kajiado temperatures), and logical reasoning from higher temperature (35 °C vs 18 °C) through lattice vibrations and electron-hole recombination to reduced power output.	Gives a clear claim with some evidence and mostly correct reasoning, with a gap in the chain of cause and effect.	States that it is hotter and gives less power without a mechanism, or says more heat gives more electricity.

Application to Kenyan contexts and scientific communication	Gives two accurate Kenyan examples linked to specific semiconductor ideas, reflects on the change in thinking since Lesson 1, and uses scientific terms correctly, with clear labelled diagrams.	Gives examples with a general link to semiconductors and uses most terms correctly; diagrams are mostly labelled.	Gives examples without linking them to the physics, uses few scientific terms, or leaves diagrams unlabelled.
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ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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